

What Are Functions?

Functions can be described as mini-programs inside your program. They're a feature of nearly every programming language. If you need to run identical instructions at three different places in a program, instead of copying and pasting the source code three times you can write the code in a function once and call the function three times. The beneficial result is a shorter and more readable program. The program is also easier to change: if you need to fix a bug or add features, you need to change your program in only one place instead of three.

All programming languages implement four features in their functions:

1. Functions have code that is run when the function is called.
2. *Arguments* (that is, values) are passed to the function when it's called. This is the input to the function, and functions can have zero or more arguments.
3. Functions return a *return value*. This is the output of the function, though some programming languages allow functions not to return anything or to return null values like `undefined` or `None`.
4. The program remembers which line of code called the function and returns to it when the function finishes its execution.

Different programming languages might have additional features, or different options for how to call functions, but they all have these four general elements. You can visually see the first three of these elements because you write them in the source code, but how does a program keep track of where the execution should return to when the function returns?

To get a better sense of the problem, create a *functionCalls.py* program that has three functions: `a()`, which calls `b()`, which calls `c()`:

Python

```
def a():  
    print('a() was called.')  
    b()
```